

Midterm Internship Report M2

Exploratory modeling of collective effects in the FCC-ee high energy booster using machine learning.

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Abstract

Modern particle accelerators such as the Future Circular Collider (FCC-ee) require increasingly precise modeling of beam dynamics in regimes where nonlinear collective effects become significant. These effects, including space-charge forces, wakefields, and beam-induced fields, introduce complex correlations in the six-dimensional phase space distribution that are computationally expensive to simulate using traditional tracking methods.

This internship focuses on the development of machine learning-based surrogate models for the efficient and physically consistent modeling of collective beam dynamics in the FCC-ee booster. In particular, the goal is to construct data-driven representations of the six-dimensional beam distribution evolution under both single-particle Hamiltonian transport and collective effects in different representation schemes such as wakefield models.

The proposed approach builds upon recent advances in operator learning and physics-informed neural networks. Neural operator architectures, such as Fourier Neural Operators, provide a natural framework for learning mappings between function spaces, making them well-suited to model the evolution of phase space densities rather than individual particle trajectories. In parallel, structure-preserving neural architectures inspired by symplectic maps (e.g. Henon-type networks) are considered to ensure consistency with the underlying Hamiltonian dynamics.

In future developments, the dynamics will be decomposed into a Hamiltonian transport operator and a collective interaction operator, allowing for efficient composition of learned evolution steps. This hybrid approach aims to provide fast, accurate, and physically interpretable surrogate models for nonlinear and collective beam dynamics, enabling improved simulation, control, and optimization of next-generation accelerator systems.

Key words: Machine learning, accelerator physics, FCC-ee, beam dynamics, collective effects, neural operators, Fourier neural operator, symplectic neural networks, operator learning, surrogate modeling.

Contents

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